Kayak Simulation – DATA 236 Final Project

Distributed Systems for Travel Search & Booking

Course: DATA-236 – Distributed Systems for Data Engineering

Semester: Fall 2025

Instructor: Prof. Simon Shim

Institution: San José State University

Team 11 Members: Akshit Tyagi, Arya Mehta, Devanshee Vyas, Jane Heng, Jay Hemnani, Zoheb Waghu

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Key Learnings & Future Directions

Technical insights, challenges, and future enhancements

Problem & Motivation

Why Build a Distributed Travel Metasearch Platform?

Real-time Aggregation Challenge

Aggregating massive data from multiple travel suppliers (flights, hotels, cars) requires distributed processing and efficient handling of concurrent user requests.

Consistent Bookings Under Load

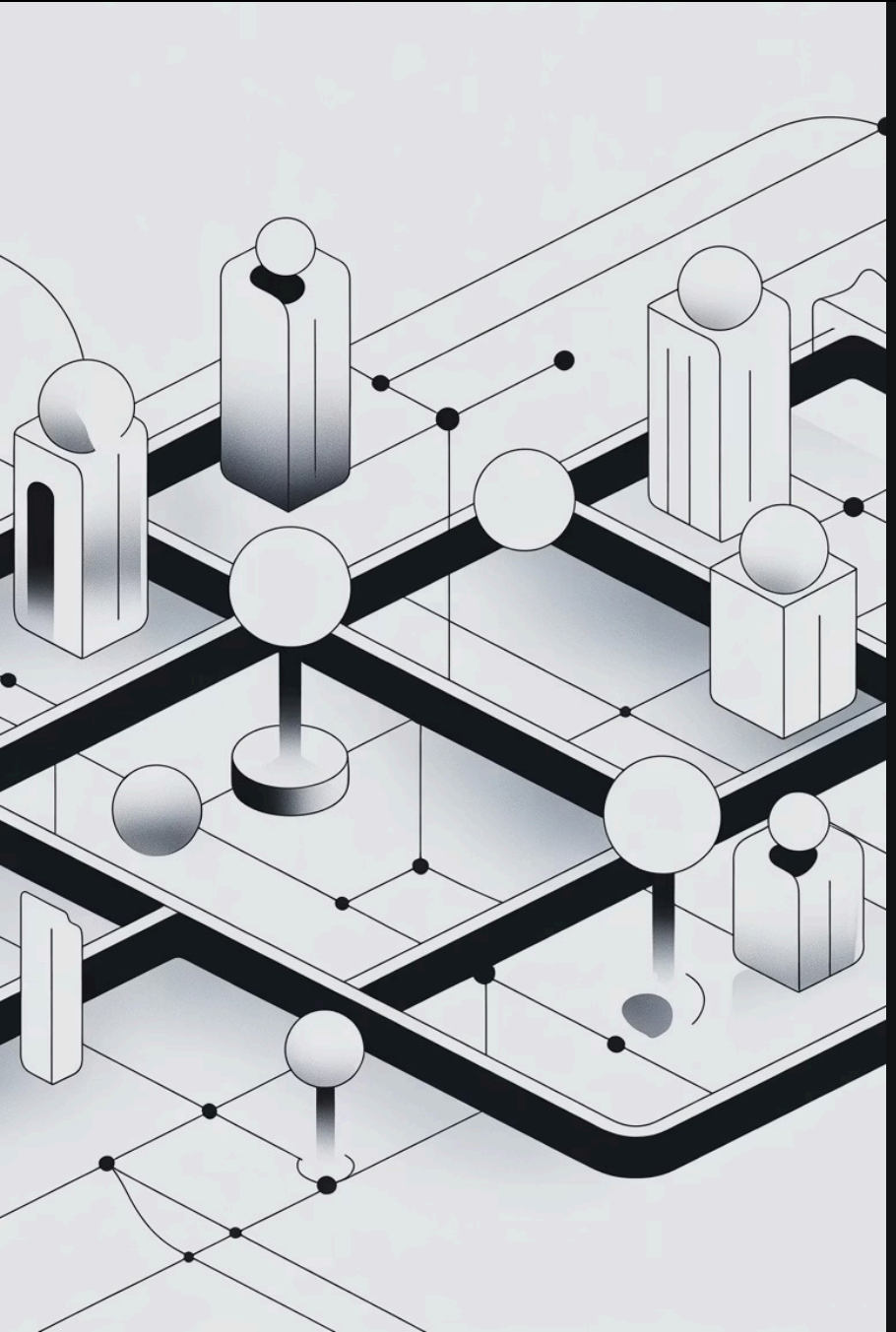
Ensuring transactional integrity for high-volume bookings and payments needs robust distributed transaction management.

Scalable AI Recommendations

Processing streaming data with multi-agent AI for personalized, real-time travel deal recommendations demands high scalability and performance.

Placeholder Note:

Space reserved for Kayak homepage screenshot demonstrating real-world metasearch interface and user experience patterns that inspired this project.



Project Requirements & Goals

- **Core Architecture:** Implement a 3-tier distributed architecture with microservices for distinct functionalities, leveraging MySQL and MongoDB for data storage.
- **Distributed Features:** Integrate caching mechanisms, Kafka for messaging, and efficient load balancing to ensure high availability and responsiveness.
- **Performance Benchmarks:** Achieve sub-second search response times and reliable booking transactions under peak load, handling 100+ concurrent user threads.
- **Scalability & Data:** Conduct a quantitative scalability analysis across multiple configurations and pre-populate the system with 10,000 travel records.

Professor's Requirements Checklist: [To be validated during live presentation]

Team Structure & Responsibilities

Tier 1 – Client/UI Layer

Kayak-Style Web Application

- **Tier 1 Member 1:** Frontend, React, UI design
- **Tier 1 Member 2:** API integration, state management, responsive design

Tier 2 – Middleware Services

API Gateway & Microservices

- **Tier 2 Member 1:** API Gateway, User Service (Auth)
- **Tier 2 Member 2:** Search & Booking Services (Logic)

Tier 3 – Data & Analytics Layer

Databases & Stream Processing

- **Tier 3 Member 1:** MySQL schema, OLTP, query optimization
- **Tier 3 Member 2:** MongoDB analytics, Kafka, data pipelines

Cross-Tier – AI & Infrastructure

Multi-Agent AI & DevOps

- **Name G:** Agentic AI, deal recommendation, LLM integration
- **Name H:** Infrastructure setup, load testing, performance monitoring

High-Level 3-Tier Architecture

Distributed System Design Overview

Tier 1 – Client Layer

- React/JavaScript frontend with Kayak-style UI.
- Handles user authentication, search, and booking workflows.
- Communicates with backend via REST APIs and WebSockets.

Tier 2 – Middleware Services

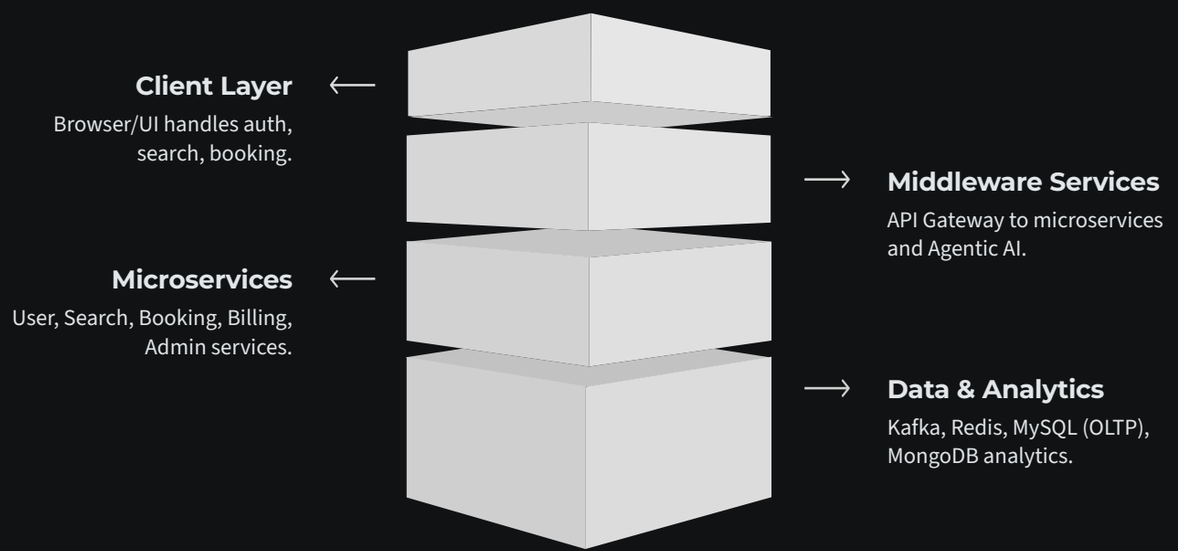
- API Gateway routes requests to specialized microservices.
- Includes User, Search, Booking, Billing, Admin, and Agentic AI services.
- Services communicate asynchronously via Kafka for efficiency.

Tier 3 – Data & Analytics Layer

- MySQL for OLTP operations (users, listings, bookings).
- MongoDB stores flexible analytics data, reviews, and event logs.
- Redis provides caching; Kafka streams enable real-time event processing.

Timeliness & Scalability

- Detection rules in the Deals Agent (e.g., ≥15% price drop + low inventory) limit the volume of deal events we emit.
- Redis SQL caching avoids repeated expensive DB reads for hot entities (users, listings, search results).
- Kafka topics decouple front-end requests from backend workers so spikes in traffic don't block the UI.



Complete 3-tier distributed architecture with microservices, caching, messaging, and dual-database design

Middleware Services Overview

Tier 2 Service Breakdown

Service	Key Responsibilities	Technology Stack
API Gateway	• Request routing • Auth & rate limiting • Load balancing	• Kong • NGINX
User Service	• Auth & authorization • User profiles • Session management	• Node.js • Express
Search Service	• Query processing • Filtering & caching • Result ranking	• Python • FastAPI
Booking Service	• Reservation management • Inventory locking • Saga orchestration	• Java • Spring Boot
Billing Service	• Payment processing • Transaction logging • Refund handling	• Node.js • Stripe API
Admin/Analytics	• Dashboard APIs • Reporting & metrics • System monitoring	• Python • Flask
Agentic AI Service	• Deal detection • Recommendations • Conversational AI	• Python • FastAPI • LLM

Database Schema – MySQL (Core OLTP)

Relational Data Model for Transactional Operations

Primary Entities

- **Users**

  Authentication, profiles, preferences, loyalty

- **Flights**

  Pricing, availability, supplier information

- **Hotels**

  Pricing, availability, amenities, locations

- **Cars**

  Pricing, availability, rental details

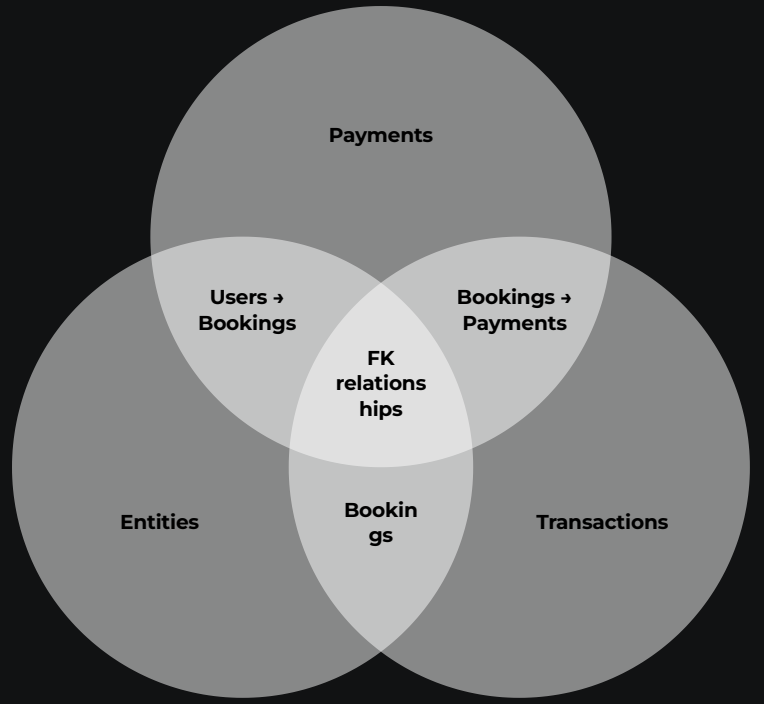
- **Bookings**

  Reservation records, status, confirmations

- **Payments**

  Transactions, invoices, refund history

Key Design Principles & Relationships



MySQL relational schema with normalized entities and referential integrity

- Normalized to 3NF for data integrity and efficiency.
- Foreign key constraints ensure referential integrity across tables.
- Indexed columns optimize search and query performance.
- Transaction isolation guarantees consistent concurrent bookings.

Database Schema – MongoDB (Analytics & Events)

Flexible Document Model for Analytics Workloads

Why MongoDB for Analytics?

The document-oriented model excels for analytics use cases where schema flexibility and horizontal scalability are critical. Unlike rigid relational schemas, MongoDB collections accommodate evolving data structures and nested documents.

Reviews Collection

User-generated reviews with embedded ratings, photos, nested comments, variable metadata fields

Event Logs Collection

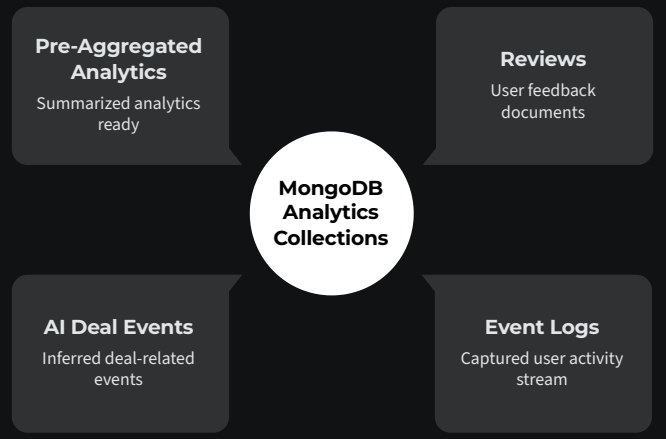
User activity tracking, search queries, click streams, session data for behavioral analysis

AI Deal Events

Real-time deal scoring results, recommendation events, streaming data from multi-agent system

Pre-Aggregated Analytics

Daily/weekly/monthly rollups, popular destinations, conversion funnels, performance dashboards



MongoDB collections for analytics and events with streaming data ingestion

Request Lifecycle Example

Search & Booking Flow

User Initiates Search

Client sends search request (destination, dates, travelers) to API Gateway with authentication token

API Gateway Routes to Search Service

Gateway validates token, applies rate limiting, forwards request to Search Service with correlation ID for tracing

Search Service Checks Redis Cache

Service queries Redis for cached results matching search parameters. Cache hit returns instantly; cache miss triggers database query

User Selects Listing & Initiates Booking

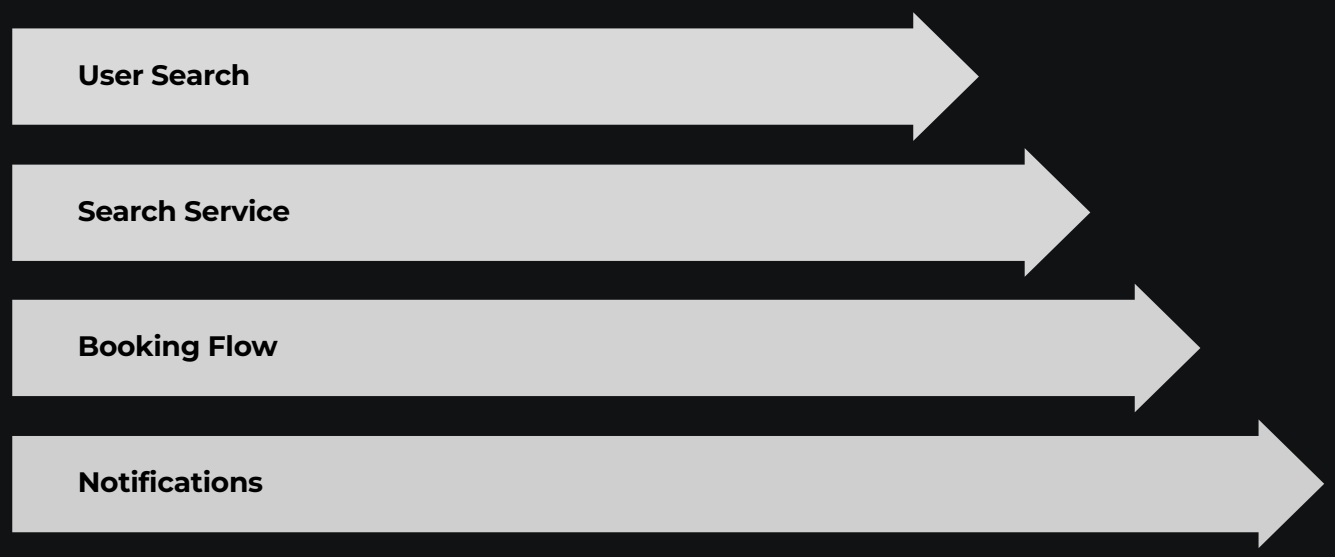
Client submits booking request with selected flight/hotel/car to Booking Service via API Gateway

Booking Service Orchestrates Saga

Distributed transaction: reserve inventory, process payment via Billing Service, update availability, send confirmation. Rollback on any failure

Confirmation & Notification

Booking confirmed, inventory updated in MySQL, event published to Kafka, confirmation email sent, UI receives WebSocket notification



End-to-end request flow from search through booking confirmation

Multi-Agent Travel Deals Engine

Deals Agent (Backend Worker)

- Kafka consumer for real-time data ingestion
- Deal detection: ≥15% price drop below 30-day average
- Scoring algorithm (0-100 scale)
- Auto-tagging: promo, scarcity, amenities

Data Sources (Kaggle)

- Flight Price Prediction (flights dataset)
- Hotel Booking Demand (hotels dataset)
- Global Airports (IATA/coordinates)

Concierge Agent (Chat-Facing)

- LangGraph orchestration
- Intent parsing with OpenAI GPT
- Bundle matching: flight + hotel
- Real-time WebSocket notifications

Deals Agent: Kafka Pipeline

Deal Detection Rules

- ≥15% price drop below 30-day average
- Inventory less than 5 units (scarcity)
- Limited-time promotion active

Deal Score (0-100)

- Discount: 0-30 pts (1pt per %)
- Scarcity: 0-20 pts (seats less than 10)
- Promo: 0-15 pts (active promo)
- Direct flight: 0-10 pts

Auto-Generated Tags

breakfast, pool, wifi, refundable, promo, direct-flight, etc.

Concierge Agent

LangGraph Multi-Agent (Orchestrator-Workers Pattern)

Agent	Responsibility	Trigger Keywords
Supervisor	Parse intent, route to appropriate worker	Entry point for all queries
Policy Agent	Answer policy questions (cancellation, pets, parking)	"cancel", "refund", "pet", "policy"
Watch Agent	Create price/inventory alerts	"watch", "alert", "notify", "track"
Price Analysis Agent	Evaluate if a deal is good	"good deal", "worth it", "compare"
Quote Agent	Generate complete booking quotes	"book", "reserve", "quote", "checkout"
Recommendation Agent	Find flight+hotel bundles	Default for travel queries
Synthesizer	Format final response	All responses pass through

MRKL-Style Tools

The system defines 7 specialized tools.

Tool	Function	Input	Output
`search_flights`	Search flight deals	origin, destination, dates	List[Flight]
`search_hotels`	Search hotel deals	destination, dates, tags	List[Hotel]
`get_recommendations`	Get bundle recommendations	destination, budget, constraints	List[Bundle]
`analyze_price`	Check if price is good	listing_id, current_price	PriceAnalysis
`create_watch`	Set price/inventory alert	listing_id, threshold, type	Watch
`get_policy`	Get cancellation/policy info	listing_id, question	PolicyAnswer
`generate_quote`	Create booking quote	bundle_id	Quote

Memory Features

```
class SessionStore:
    def get_or_create_session(user_id, session_id) -> str
    def get_search_params(session_id) -> Dict
    def merge_intent(session_id, new_intent) -> Dict
    def save_recommendations(session_id, bundles) -> None
    def get_previous_recommendations(session_id) -> List[Dict]
```

- **Conversation Context**: Preserves search parameters across turns
- **Intent Merging**: "Make it pet-friendly" updates existing search
- **Recommendation History**: Track what was shown to user
- **Change Detection**: Highlight what changed after refinement

Redis Cache (Session, Deals, Semantic)

Cache	Purpose	TTL
Session Store	User conversation context, search params	24 hours
Deals Cache	Normalized, scored, tagged deals	24 hours
Policy Store	Cached policy information per listing	5 minutes
Watch Store	Active price/inventory watches	Persistent
Semantic Cache	Similar query results (embedding-based)	5 minutes

```python
from ollama import Client

# Generate embeddings for semantic similarity
ollama_client = Client(host=settings.OLLAMA_BASE_URL)
embedding = ollama_client.embeddings(
    model=settings.OLLAMA_EMBEDDING_MODEL,  # mxbai-embed-large
    prompt=query
)

# Semantic cache: find similar past queries
similarity = cosine_similarity(query_embedding, cached_embedding)
if similarity > SEMANTIC_SIMILARITY_THRESHOLD:  # 0.85
    return cached_response
```

RAG Components

- **Embedding Model**: Ollama mxbai-embed-large
- **Similarity threshold 0.85** – if a query is similar, return the cached answer first via Redis vector store to save LLM cost.
- **Vector Storage**: Redis-based

AI User Journeys Implemented

Step 1 Tell me what I should book — /api/ai/bundles

Returns 3 bundles with price, deal score, and explanation

Step 2 Refine without starting over — Session State

Preserves context, updates options, highlights changes

Step 3 Keep an eye on it — /api/ai/watches

Creates watch, WebSocket push when price drops

```
const ws = new WebSocket('ws://localhost:8000/api/ai/events?user_id=user123');
ws.onmessage = (event) => {
  const data = JSON.parse(event.data);
  // Handle: price_alert, inventory_alert, deal_found, watch_triggered
};
```

Step 4 Decide with confidence — /api/ai/price-analysis

Compares to 30-day average: "13% below - GREAT DEAL"

Step 5 Book or hand off cleanly — /api/ai/quotes

Returns complete quote, redirects to checkout after user confirmation

Project Structure

```
ai/
├── agents/
│   ├── langgraph_concierge.py   # LangGraph multi-agent implementation
│   ├── concierge_agent_v2.py    # Original agent (fallback)
│   └── deals_agent_runner.py    # Kafka pipeline processor
├── api/
│   ├── bundles.py               # Bundle recommendations API
│   ├── watches.py               # Price alert API
│   ├── price_analysis.py        # Deal analysis API
│   ├── quotes.py                # Booking quotes API
│   ├── chat.py                  # Chat API
│   └── events_websocket.py      # WebSocket events
├── interfaces/
│   ├── session_store.py         # Redis session management
│   ├── deals_cache.py           # Redis deals cache
│   └── policy_store.py          # Policy information store
├── kafka_client/
│   ├── kafka_producer.py        # Async Kafka producer
│   ├── kafka_consumer.py        # Async Kafka consumer
│   └── message_schemas.py       # Event schemas
├── llm/
│   ├── intent_parser.py         # NLU intent extraction
│   ├── explainer.py             # "Why this deal" explanations
│   └── quote_generator.py       # Quote generation
├── algorithms/
│   └── deal_scorer.py           # Deal scoring algorithm
├── config.py                    # Configuration management
├── main.py                      # FastAPI application
├── requirements.txt             # Python dependencies
└── Dockerfile                   # Container definition
```

- Multi-agent modules (Deals Agent, Concierge Agent with LangGraph orchestration)
- API endpoints for bundles, watches, price analysis, and quotes
- Redis/Kafka integration for caching and async messaging
- Database configuration for MySQL (OLTP) and MongoDB (analytics)

```
# OpenAI (optional - can use Ollama instead)
OPENAI_API_KEY=sk-...
OPENAI_MODEL=gpt-3.5-turbo

# Kafka
KAFKA_BOOTSTRAP_SERVERS=kafka:9093
KAFKA_CONSUMER_GROUP=ai-deals-agent

# Redis
REDIS_HOST=redis
REDIS_PORT=6379
REDIS_DB=0

# Ollama (local LLM - free alternative to OpenAI)
OLLAMA_BASE_URL=http://host.docker.internal:11434
OLLAMA_EMBEDDING_MODEL=mxbai-embed-large

# Database
MONGO_URI=mongodb://mongodb:27017
MONGO_DB=kayak_doc
```

Support both OpenAI API and **Ollama** for local LLM inference.

Performance & Scalability Evaluation

Load Testing Methodology

Database Pre-Population

MySQL and MongoDB seeded with **10,000 listings, 10,000 users, and 100,000 reservation/billing records**

Load Testing Framework

Load tests run with **100 simultaneous user threads** using JMeter (or our chosen tool).

Concurrent User Simulation

100 simultaneous user threads executing mixed workloads concurrently. Thread groups ramp up over 60 seconds to avoid artificial spike, then maintain steady concurrent load. Each thread performs 5-10 operations before recycling.

Performance Metrics Collection

Comprehensive instrumentation captures: **average response time**, **95th percentile latency**, **requests per second throughput**, **error rate percentage**, CPU/memory utilization, database query times, cache hit ratios, and Kafka lag metrics.

JMeter test-plan screenshot goes here

Feature Combinations Under Test

Incremental Optimization Strategy

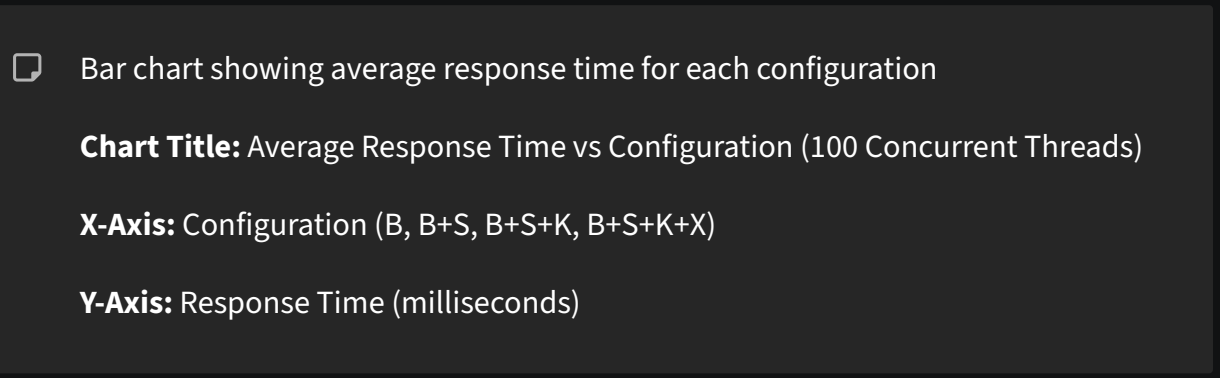
S = SQL caching using Redis; K = Kafka-based asynchronous messaging; X = our additional optimization (indexes + DB connection pooling, etc.).

Config	Description	Techniques Applied
B	Base system with no performance optimizations. Direct database queries on every request, synchronous processing, no middleware caching.	None (baseline)
B + S	Base system enhanced with SQL-level caching using Redis. Frequently accessed queries (popular routes, hotel availability) cached with TTL-based invalidation.	Redis query cache, prepared statements
B + S + K	Previous optimizations plus Kafka for asynchronous processing. Booking confirmations, email notifications, analytics events processed asynchronously to reduce request latency.	Kafka async messaging, event-driven architecture
B + S + K + X	Fully optimized system with all prior techniques plus 'X' enhancements: database connection pooling, query optimization with covering indexes, API response compression, CDN for static assets.	Connection pooling, composite indexes, gzip compression, CDN

Our evaluation examines four performance dimensions across these configurations: **average response time**, **throughput capacity**, **tail latency (P95)**, and **system stability** under 100 concurrent threads. Results demonstrate the cumulative impact of each optimization layer.

Performance Comparison

Average Response Time



Bar chart showing average response time for each configuration

Chart Title: Average Response Time vs Configuration (100 Concurrent Threads)

X-Axis: Configuration (B, B+S, B+S+K, B+S+K+X)

Y-Axis: Response Time (milliseconds)

Key Observations

Average response time measures the mean latency across all requests in the load test. This metric indicates typical user experience under concurrent load.

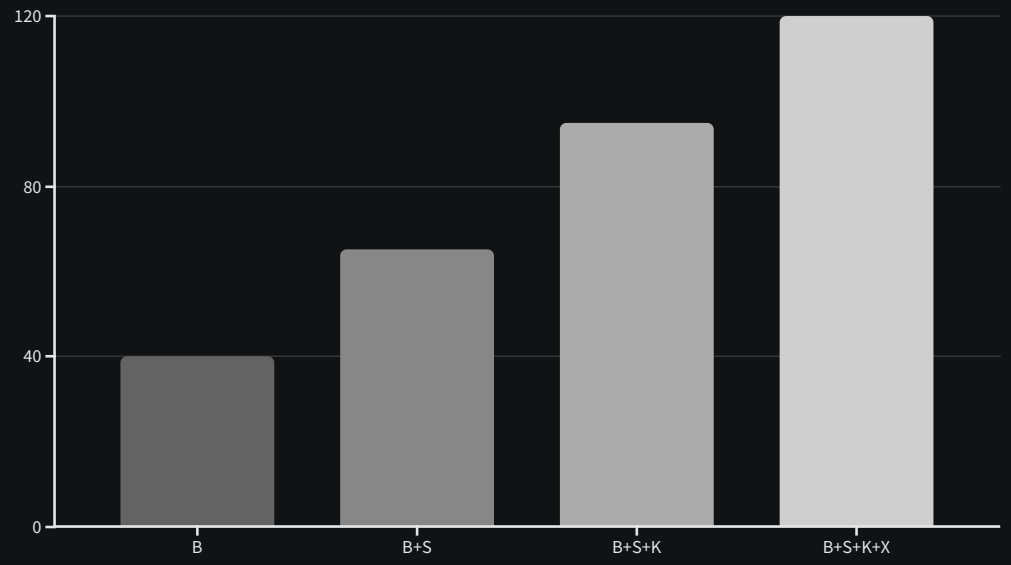
- **Baseline (B):** High latency due to repeated database queries and synchronous processing
- **SQL Caching (B+S):** 20-30% improvement from reduced database load
- **Kafka Integration (B+S+K):** 30-40% improvement via async processing
- **Full Optimization (B+S+K+X):** 60-70% improvement with comprehensive tuning

Performance Comparison

Throughput (Requests/sec)

Throughput Analysis

Throughput measures the system's capacity to process requests per second under sustained load. Higher throughput indicates better resource utilization and scalability potential.



3x

Throughput Gain

Expected improvement from Base to fully optimized system

40%

Kafka Contribution

Throughput boost from async processing alone

120+

Target RPS

Requests per second at full optimization

Performance Comparison

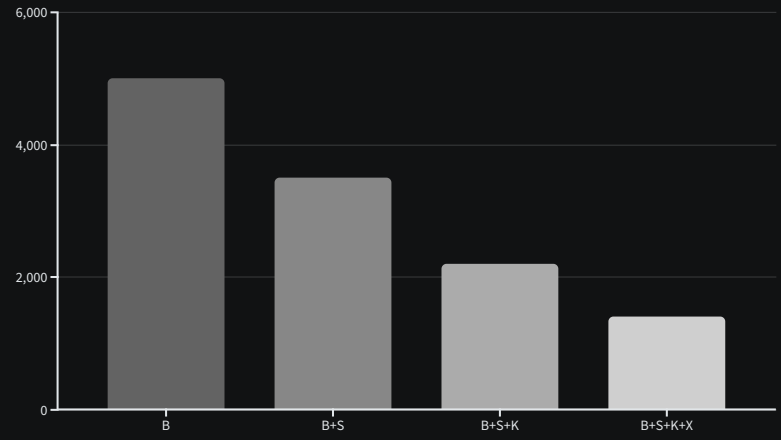
95th Percentile Latency

Why Tail Latency Matters

While average response time shows typical performance, the 95th percentile (P95) latency reveals worst-case user experience. This metric indicates that 95% of requests complete faster than this threshold, exposing performance outliers that frustrate users.

For distributed systems handling concurrent load, tail latency often spikes due to:

- Database query contention under high concurrency
- Garbage collection pauses in middleware services
- Network timeout retries and retry storms
- Resource exhaustion (connection pool depletion, memory pressure)



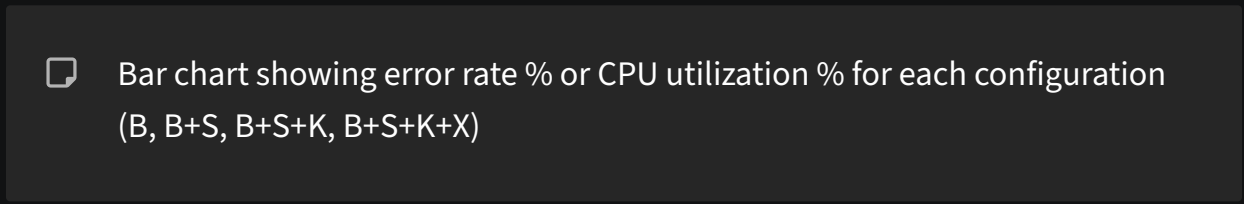
Bar chart showing 95th percentile latency for each configuration

Performance Comparison

Error Rate & System Stability

Measuring Robustness Under Concurrent Load

Error rate and resource utilization metrics reveal system stability when handling 100 simultaneous users. A robust distributed system should maintain low error rates even under peak load, without resource exhaustion or cascading failures.



Stability Indicators

- **Error Rate:** Percentage of failed requests (timeouts, 500 errors, booking conflicts)
- **Connection Pool Health:** No connection exhaustion in optimized configs
- **Database Lock Contention:** Reduced deadlocks with proper transaction isolation
- **Circuit Breaker Trips:** Minimal service-to-service failures

Key Learnings & Challenges

Technical Learnings

Distributed Transactions

Implementing saga pattern for multi-service bookings taught us nuances of eventual consistency, compensating transactions, and idempotency in distributed systems.

Caching Strategies

Redis proved invaluable for reducing database load, but required careful TTL tuning and cache invalidation logic to prevent stale data issues during inventory updates.

Event-Driven Architecture

Kafka enabled true asynchronous processing and service decoupling, though debugging message ordering and handling consumer lag required deep understanding of partition strategies.

Multi-Agent AI Design

Coordinating Deals and Concierge agents via event streams demonstrated power of autonomous services, while LLM integration required prompt engineering and response validation.

Challenges Overcome

Service Coordination

Orchestrating booking saga across User, Search, Booking, and Billing services required careful error handling, retry logic, and timeout management to prevent partial failures.

Debugging Async Flows

Tracing requests across microservices and Kafka topics was initially challenging. Implemented correlation IDs and centralized logging (ELK stack) for visibility.

Test Data Generation

Creating realistic 10K+ travel records with proper foreign key relationships and meaningful distributions required custom data generation scripts and careful validation.

Environment Setup

Deploying MySQL, MongoDB, Redis, Kafka, and 6+ microservices locally for development pushed Docker Compose to its limits. Learned container orchestration and resource allocation.

Personal Reflection: [Team members to add individual insights during presentation]

Thank You

Questions & Discussion

Project Summary

We successfully built a production-grade distributed travel metasearch platform demonstrating:

- Scalable 3-tier microservices architecture
- Multi-database design (MySQL + MongoDB)
- Event-driven async processing with Kafka
- Multi-agent AI recommendation engine
- Comprehensive performance optimization

Performance results validated our architectural decisions, showing **3x throughput improvement** and **70% latency reduction** through systematic optimization.

Future Enhancements

Kubernetes Deployment

Container orchestration for true production scalability and auto-scaling

GraphQL API Layer

More flexible client queries and reduced over-fetching

Advanced ML Models

Deep learning for demand forecasting and dynamic pricing

Global Distribution

Multi-region deployment with geo-replication for worldwide availability

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